INTRODUCTION TO FOURIER SERIES AND THEIR CONVERGENCE IN L^p

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Abstract. We first develop some basic theory of Fourier series, and then we investigate the convergence of Fourier series with respect to the L^p norms. We give counterexamples in the case of $p = 1$ and $p = \infty$, and bound the growth of the Fourier partial sums in those cases. We end with a brief discussion of the Carleson-Hunt theorem.

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1. Introduction

The field of harmonic analysis, or for our purposes, Fourier analysis, has many applications in PDEs and physics, but the introduction of Fourier series also revealed many deficiencies in the mathematical rigor of the work being done at the time - there were many non-pathological examples of Fourier series not converging, and there were issues with trying to integrate and differentiate Fourier series.

Riemann realized that a more robust theory of integration was needed attempted to remedy this by developing the Riemann integral, but unfortunately Riemann integration does not have strong enough convergence theorems to swap sums and integrals. Eventually, Lebesgue's thesis provided a theory of integration with convergence theorems with mild conditions, and the theory of L^p spaces put Fourier analysis on solid ground. In a way, this is a rather fitting topic for a final project to highlight the importance of measure theory beyond artificial examples like the Dirichlet function.

Of course, this means we are very interested in the convergence properties of the Fourier series of various functions. We mainly explore the convergence of Fourier partial sums in L^p norms, discussing the result that $s_N(f) \to f$ in L^p norm for $1 < p < \infty$ where $s_N(f)$ is the Nth Fourier partial sum of f. We then look at why convergence fails in the cases $p = 1$ and $p = \infty$, and we prove that

$$\|s_N\|_{op} = \frac{4}{\pi^2} \log N + O(1)$$

where $\|\cdot\|_{op}$ denotes the L^1 operator norm, and

$$\lim_{n\to\infty} \frac{\|s_N(f)\|_{L^\infty(\mathbb{T})}}{\log N} = 0$$

At the end, we also briefly touch on the Carleson-Hunt theorem and discuss its importance.

2. Basic Fourier Analysis

For the purpose of this paper, I will mostly only assume the content from Math 245A, as well as undergraduate analysis, and not assume any prerequisite knowledge about Fourier series. I myself did not encounter Fourier analysis while taking the undergraduate real analysis sequence at NYU so it will also be beneficial for me to explain some of the basics.

2.1. Motivation.

To motivate the definition of Fourier series, we first look at a few examples in physics and discuss them non-rigorously.

Example 2.1 (Driven Harmonic Oscillator)**.** Consider a spring-mass system. In the case of simple harmonic motion, its equation of motion is given by

$$m\ddot{x} + kx = 0$$

The general form of the solution is given by $x(t) = A\sin(\omega t) + B\cos(\omega t)$ where $\omega = \sqrt{k/m}$. Depending on initial conditions, this corresponds pulling the mass to a certain initial position, releasing it, and letting it oscillate. Note that if we allow the use of complex numbeers, we can also rewrite the above solution as the real part of $Ae^{-i\omega t} + Be^{i\omega t}$.

However, we may also want to consider situations in which we apply an external force on the system. For the sake of simplicity suppose this force depends only on time and denote it by $f(t)$. Then our equation of motion becomes

$$m\ddot{x} + kx = f(t)$$

To solve an inhomogeneous ODE we must find a particular solution, but this becomes difficult for an arbitrary driving force $f(t)$. If we restrict ourselves to oscillating forces of the form $f(t) = \sin(at)$ $(a \neq \omega)$, we can use an ansatz to show that a particular solution is given by $C\sin(at)$ for some constant C. (It is slightly more complicated when $a = \omega$ but not by much.) We can also do the same thing if the driving force is a cosine.

Since this ODE is linear, a particular solution for the RHS $f(t) = g(t) + h(t)$ is the sum of some particular solution for the RHS $g(t)$ and some particular solution for the RHS $h(t)$. This means we can solve the driven harmonic oscillator for any force that we can write as a finite linear combination of sines and cosines! The natural question to ask now is: does this extend to infinite linear combinations? And a little bit of wishful thinking may lead us to ask: is it possible to write any arbitrary function $f(t)$ as a sum of sines and cosines? This is exactly a Fourier series!

We now look at a classic example in which Fourier series arise as solutions.

Example 2.2 (Heat Equation)**.** The heat equation in 1 dimension is given by

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

This was in fact the original example that motivated Fourier to develop the notion of Fourier series. The equation describes how heat permeates through a region. To get specific solutions we also need to impose boundary conditions and initial conditions, for example

$$\begin{cases} u(x,0) = f(x) & 0 \le x \le L \\ u(0,t) = u(L,t) = 0 & t > 0 \end{cases}$$

which describes heat diffusing through a rod of length L and $f(x)$ describes the initial state.

We may look for solutions by separation of variables, that is, we look for solutions of the form $u(x,t) = X(x)T(t)$. Plugging this into the equation and rearranging, we get

$$\frac{T'}{T} = \frac{X''}{X}$$

Since the LHS only depends on t and the RHS only depends on x, they must each be equal to some constant, say $-\lambda$, for $\lambda \in \mathbb{R}$. We then get ODEs in t and x separately. I'll omit the details here, but taking $\lambda \leq 0$ only gives the trivial solution 0. Taking $\lambda > 0$, we get solutions

$$T(t) = Ae^{-\lambda t},\ X(x) = Be^{i\sqrt{\lambda}x} + Ce^{-i\sqrt{\lambda}x}$$

Imposing the boundary conditions, we can see λ can only take on a discrete number of values, given by

$$\lambda_n = \frac{n^2\pi^2}{L^2},\ n \in \mathbb{N}^+$$

We can then index T and X with this discrete value n. Imposing the same boundary conditions on X, we see that $X_n(x) = \sin(\sqrt{\lambda_n}x)$. Finally, when we try to apply the initial condition $u(x,0) = f(x)$, we see that $u_n(x,t) = X_n(x)T_n(t)$ is a solution to the heat equation for $f(x)$ of the form $\sin(\sqrt{\lambda_n}x)$. Like the previous example, we can then try to naïvely construct a full solution of the form

$$u(x,t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi}{L}x\right) e^{-\frac{n^2\pi^2}{L^2}t}$$

for initial conditions of the form

$$f(x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi}{L}x\right)$$

Again this naturally begs the question, which initial state functions can be written in such a form, as a Fourier series?

These examples illuminate the usefulness of the idea of a series of trigonometric functions, but as we mentioned in the introduction, many issues arise with the convergence properties of these series.

Example 2.3. Consider $f(x) = x$. Assuming for now we have a method of expanding f in terms of trigonometric functions. The Fourier series of f then turns out to be

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{4}{n}\sin\left(\frac{nx}{2}\right)$$

Recall sine functions are 2π-periodic, so a natural interval to examine is $[0, 2\pi)$. Fortunately this series converges to x everywhere in this interval except at 0. However, if we try to use term by term differentiation, we get

$$\sum_{n=1}^{\infty}(-1)^{n+1}\cdot 2\cos\left(\frac{nx}{2}\right)$$

which actually converges nowhere in $[0, 2\pi)$. This failure is caused by the lack of uniform convergence, but more fundamentally due to the fact that x is not periodic at all, so when we try to consider x as a periodic function on $[0, 2\pi)$, we essentially get a jump continuity between the ends.

This shows that we not only have to handle the series carefully, but we also have to be careful in choosing what kinds of functions to represent as Fourier series - there is clearly some kind of issue when it comes to trying to represent non-periodic functions as Fourier series.

2.2. **Functions on $\mathbb{T}$.** We now start rigorously defining Fourier series, closely following Katznelson's book.

Definition 2.4. Let $\mathbb{R}$ and $\mathbb{Z}$ be additive groups. Then we define the 1 dimensional torus, or the unit circle, by $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$.

Remark 2.5. This definition of $\mathbb{T}$ is useful for us since it allows to talk about 2π-periodic functions as functions on a compact space, and develop measure theory on this space, but it can also be identified with the unit circle $\partial\mathbb{D}$ in the complex plane, which allows us to also talk about holomorphicity, and holomorphic functions on the unit disk are very well studied!

Then, we have a homeomorphism between $\mathbb{T}$ and $[0, 2\pi]/\sim$ where $0 \sim 2\pi$, so we can identify functions on $\mathbb{T}$ with 2π-periodic functions on $\mathbb{R}$. This means topological properties, differentiability, and so on carry over. We define integration as follows.

Definition 2.6 (Integration on $\mathbb{T}$)**.**

$$\int_{\mathbb{T}} f(t)\, dt := \int_0^{2\pi} f(x)\, dx$$

Then the Lebesgue measure on $\mathbb{T}$ can essentially be identified with a restriction of the Lebesgue measure on $\mathbb{R}$ to $[0, 2\pi)$. An important property that carries over is the fact that the measure is also translation invariant,so

Proposition 2.7 (Translation Invariance of $\mathbb{T}$)**.**

$$\int_{\mathbb{T}} f(t - t_0)\, dt = \int_{\mathbb{T}} f(t)\, dt, \quad t_0 \in \mathbb{T}$$

Now we can similarly define the L^p spaces on $\mathbb{T}$, but we add in a normalization factor of $1/2\pi$ to compensate for the fact that $\mathbb{T}$ is identified with $[0, 2\pi)$.

Definition 2.8 (L^p spaces on $\mathbb{T}$)**.** For $1 \le p < \infty$ we define the L^p norm of a measurable function $f : \mathbb{T} \to \mathbb{C}$ by

$$\|f\|_{L^p(\mathbb{T})} := \left(\frac{1}{2\pi} \int_{\mathbb{T}} |f|^p \right)^{1/p}$$

We then define $L^p(\mathbb{T})$ to be the collection of all the (equivalence classes of) measurable functions with finite L^p norm. (As usual, we declare $f \sim g$ iff $f = g$ almost everywhere.)

We define the L^∞ norm by

$$\|f\|_{L^\infty(\mathbb{T})} := \operatorname*{ess\,sup}_{t \in \mathbb{T}} |f(t)| := \inf\{C \ge 0 : \mu(\{t \in \mathbb{T} : |f(t)| > C\}) = 0\}$$

where μ is the Lebesgue measure on $\mathbb{T}$. Then $L^\infty(\mathbb{T})$ is similarly taken to be the collection of (equivalence classes of) functions with finite L^∞ norm.

It is then a standard fact that the L^p spaces are complete, and the L^p spaces are Banach spaces. We give some useful facts about L^p spaces on $\mathbb{T}$.

Proposition 2.9. *If $f_n \in L^p(\mathbb{T})$ converges uniformly to $f \in L^p(\mathbb{T})$, then $f_n \to f$ in L^p norm as well.*

Proof. Since $f_n \to f$ uniformly we have that $\|f_n - f\|_\infty \to 0$, so it is bounded. We have

$$\begin{aligned}
\|f_n - f\|_{L^p(\mathbb{T})} &= \left(\frac{1}{2\pi} \int_{\mathbb{T}} |f_n - f|^p \right)^{1/p} \\
&\le \left(\frac{1}{2\pi} \|f_n - f\|_\infty^p \mu(\mathbb{T}) \right)^{1/p} \\
&= \|f_n - f\|_\infty \to 0
\end{aligned}$$

□

Proposition 2.10. *$C(\mathbb{T}) \subseteq L^p(\mathbb{T})$ for all $1 \le p < \infty$ where $C(\mathbb{T}$ is the set of continuous functions on $\mathbb{T}$.*

Proof. $\mathbb{T}$ is compact, so any $f \in C(\mathbb{T})$ is bounded. Then a simple computation shows $\|f\|_{L^p(\mathbb{T})} < \infty$. $\square$

The above statement happens to be true for $p = \infty$, but the main purpose of this proposition is so the statement "continuous functions are dense in L^p spaces" makes sense, but this statement is not true for $p = \infty$.

Proposition 2.11. $L^p(\mathbb{T}) \subseteq L^1(\mathbb{T})$

Proof. Let $f \in L^p(\mathbb{T})$. Then we have

$$\int_{\mathbb{T}} |f(t)|\, dt = \int_{\{t\in\mathbb{T}:|f(t)|\leq 1\}} |f(t)|\, dt + \int_{\{t\in\mathbb{T}:|f(t)|>1\}} |f(t)|\, dt$$

We analyze each integral separately. We have

$$\int_{\{t\in\mathbb{T}:|f(t)|\leq 1\}} |f(t)|\, dt \leq \int_{\{t\in\mathbb{T}:|f(t)|\leq 1\}} 1\, dt \leq \int_{\mathbb{T}} 1\, dt = \mu(\mathbb{T}) < \infty$$

and

$$\int_{\{t\in\mathbb{T}:|f(t)|>1\}} |f(t)|\, dt \leq \int_{\{t\in\mathbb{T}:|f(t)|>1\}} |f(t)|^p\, dt \leq \int_{\mathbb{T}} |f(t)|^p\, dt = \|f\|_{L^p(\mathbb{T})}^p < \infty$$

Thus $\int_{\mathbb{T}} |f| < \infty$ and $f \in L^1(\mathbb{T})$. $\square$

We will only explicitly define Fourier series for L^1 functions, but this allows us to talk about the Fourier series of L^p functions as well.

2.3. **Fourier Series.** Now that we have a few basic notions of convergence for functions on $\mathbb{T}$, we can define trigonometric series and start talking about their convergence.

Definition 2.12 (Trigonometric polynomial)**.** A trigonometric polynomial on $\mathbb{T}$ is an expression of the form

$$P(t) = \sum_{n=-N}^{N} a_n e^{int}$$

Although we do not formally know the convergence behavior as $N \to \infty$, we can still formally define the infinite sum.

Definition 2.13 (Trigonometric series)**.** A trigonometric series is a formal expression of the form

$$\sum_{n=-\infty}^{\infty} a_n e^{int}$$

Note that by choosing $a_{-n} = -a_n$ and choosing $a_n \in \mathbb{R}$, we can reduce a trigonometric polynomial into a sum of sines, which is very much reminiscent of the series we obtained when examining the heat equation. But before we can ask whether these series converge, we should first ask whether there is any hope that we can write most functions as a series of this form.

Is there any reason to choose the set $\{e^{int}\}_{n\in\mathbb{Z}}$ as the "basis" for our series? Yes, the first reason is that they span "almost every" function in the L^p spaces.

Theorem 2.14. *Trigonometric polynomials are dense in* $L^p(\mathbb{T})$.

This says that we can approximate any function in $L^p(\mathbb{T})$ arbitrarily well by triogonometric polynomials. The proof can be broken up into two steps: we first show that the trigonometric polynomials are dense in the continuous functions on $\mathbb{T}$, and then show the continuous functions are dense in $L^p(\mathbb{T})$. To prove the first part, we use the Stone-Weierstrass theorem.

Definition 2.15. Let $C(X)$ denote the set of continuous functions on X equipped with the uniform norm. A subset $\mathcal{A} \subseteq C(X)$ is set to separate points if for all $x, y \in X$ there exists $f \in \mathcal{A}$ such that $f(x) \neq f(y)$

Definition 2.16. A subset $\mathcal{A} \subseteq C(X)$ is a subalgebra if it is a vector subspace of $C(X)$ and is closed under function multiplication, that is, $fg \in \mathcal{A}$ whenever $f, g \in \mathcal{A}$.

Theorem 2.17 ((Complex) Stone-Weierstrass)**.** *Let X be a compact Hausdorff space and $\mathcal{A}$ a subalgebra of $C(X, \mathbb{C})$ which contains a constant function, is closed under conjugation, and separates points. Then $\mathcal{A}$ is dense in $C(X, \mathbb{C})$.*

We will omit the proof for the more topological theorems in the interest of time, since they are less relevant to the measure theory aspect of the course. (See Folland for proofs.) The next step is to show the continuous functions are dense in $L^p(\mathbb{T})$.

Recall Lusin's Theorem: (we take the following version from Folland)

Theorem 2.18 (Lusin)**.** *Let $f : [a, b] \to \mathbb{C}$ be measurable and $\varepsilon > 0$. Then there exists a compact $E \subseteq [a, b]$ such that $\mu([a, b] \setminus E) \leq \varepsilon$ and $f|_E$ is continuous.*

Remark 2.19. This version is an easy consequence of the statement in Tao's Introduction to measure theory, which says a measurable function on $\mathbb{R}$ is continuous after removing a measurable set of arbitrary measure. We can always restrict our attention to a subset $[a, b]$. Then, we remove the same measurable set. The remaining set on which the function is continuous is measurable, so we can approximate it within by closed sets (which are compact since they lie inside a bounded set). Then the above version of Lusin's theorem follows.

We get a continuous function at the cost of removing a small set. Fortunately, we can re-extend our function by the following result:

Theorem 2.20 (Tietze Extension Theorem)**.** *Let X be a normal space. If $A \subseteq X$ is closed, and $f \in C(A, [a, b])$, then there exists $F \in C(X, [a, b])$ such that $F|_A = f$.*

Again this is a purely topological theorem, so refer to Folland for the proof. Note that this says that the extended function will have the same bounds as the original function. Putting the previous two theorems together, we get our desired result.

Lemma 2.21. *The set of continuous functions $C(\mathbb{T}, \mathbb{C})$ is dense in $L^p(\mathbb{T})$.*

Proof. Recall our identification of $\mathbb{T}$ with $[0, 2\pi)$. Let $f \in L^p([0, 2\pi))$. We have that $f\mathbf{1}_{|f|\leq n} \to f$ uniformly, so it also converges in L^p norm by Proposition 2.9. Then we can pick $N \in \mathbb{N}$ such that $\|f\mathbf{1}_{|f|\leq N} - f\|_{L^p([0,2\pi))} \leq \varepsilon/2$. We denote $f_N := f\mathbf{1}_{|f|\leq N}$.

Note that our domain of definition being half-open does not affect the proof of the theorem, as we explained in the remark above, so we can still apply Lusin's theorem. Let $\varepsilon > 0$. Since f_N is continuous, it is measurable, and by Lusin's theorem we can find a compact set $E \subseteq [0, 2\pi)$ with $\mu([0, 2\pi) \setminus E) \leq 2^{-p}\varepsilon^p(2N)^{-p}$ such that $f_N|_E$ is continuous. It also bounded by some number M since E is compact.

Now all metric spaces is normal, so $[0, 2\pi)$ is normal. By the Tietze extension theorem we can extend $f_N|_E$ back to a continuous function $F : [0, 2\pi) \to \mathbb{R}$ which is also bounded by M. Then we have

$$\begin{aligned}\|f_N - F\|_{L^p([0,2\pi))} &= \left(\int_{[0,2\pi)\setminus E} |f_N - F|^p\right)^{1/p} \\ &\leq \left(\int_{[0,2\pi)\setminus E} (|f_N| + |F|)^p\right)^{1/p}\end{aligned}$$

$$\leq \left[(2N)^p \mu([0,2\pi) \setminus E)\right]^{1/p}$$
$$\leq \varepsilon/2$$

Finally, by triangle inequality, we have $\|f - F\|_{L^p([0,2\pi))} \leq \varepsilon$, which implies $\|f - F\|_{L^p(\mathbb{T})} \leq \varepsilon/(2\pi)^{1/p}$. Hence $C(\mathbb{T}, \mathbb{C})$ is dense in $L^p(\mathbb{T})$. □

Finally we can prove the full result:

Proof of Theorem 2.12. Let $\mathcal{A}$ be the algebra generated by $\{e^{inx}\}_{n\in\mathbb{Z}}$. Then $\mathcal{A}$ is exactly the set of trigonometric polynomials. Then $\mathcal{A}$ contains a constant function $e^{i\cdot 0\cdot t}$ and $\overline{e^{inx}} = e^{-inx}$ so it is closed under conjugation. e^{it} maps $\mathbb{T}$ bijectively to the unit circle $\partial\mathbb{D}$, so $\mathcal{A}$ separates points. Thus by Stone-Weierstrass we have that $\mathcal{A}$ is dense in $C(\mathbb{T}, \mathbb{C})$. Now by our previous lemma we have that $C(\mathbb{T}, \mathbb{C})$ is dense in $L^p(\mathbb{T})$. Thus $\mathcal{A}$ is dense in $L^p(\mathbb{T})$. □

This tell us that it is at least a reasonable idea to use trigonometric series to represent functions, since at least trigonometric polynomials can approximate functions arbitrarily well (although this is not quite enough to call it a basis). The second reason we choose to use the set $\{e^{inx}\}_{n\in\mathbb{Z}}$ is because of the following property:

Proposition 2.22.
$$\frac{1}{2\pi}\int_{\mathbb{T}} e^{int}e^{-imt} = \delta_{mn}$$
where δ_{mn} is the Kronecker delta.

This immediately gives us a way of extracting coefficients out of trigonometric polynomials:

Proposition 2.23. *Let*
$$P(t) = \sum_{n=-N}^{N} a_n e^{int}$$
Then
$$a_m = \frac{1}{2\pi}\int_{\mathbb{T}} P(t)e^{-imt}\,dt$$

Remark 2.24. Note that this is very reminiscent of an inner product. In fact, if we define $\langle f, g\rangle = \frac{1}{2\pi}\int_{\mathbb{T}} \overline{f}g$ we can show that it makes $L^2(\mathbb{T})$ into a HIlbert space, where $\{e^{inx}\}_{n\in\mathbb{Z}}$ is an orthogonal basis. Unfortunately this is not true for other L^p spaces, and we want to look at convergence in L^p for all p, so we will not focus on the Hilbert space structure of $L^2(\mathbb{T})$.

Now this motivates us to define the Fourier series of an arbitrary function.

Definition 2.25 (Fourier coefficients)**.** Let $f \in L^1(\mathbb{T})$. We define the nth Fourier coefficient of f by
$$\hat{f}(n) := \frac{1}{2\pi}\int_{\mathbb{T}} f(t)e^{int}\,dt$$

Definition 2.26 (Fourier series)**.** The Fourier series is the formal expression of the form
$$\sum_{n=-\infty}^{\infty} \hat{f}(n)e^{int}$$
which we denote by $s(f)$. We denote the partial sums by
$$s_N(f)(t) := \sum_{n=-N}^{N} \hat{f}(n)e^{int}$$

3. L^p Convergence of Fourier Series

Now we are ready to talk about how the Fourier series of functions converge. It might seem natural to investigate the pointwise convergence of Fourier series first, but this turns out to be the most difficult to show, and we leave its discussion until the end. We start with convergence in the L^p norms.

3.1. Case $1 < p < \infty$.

It turns out that for most values of p we have a very simple statement:

Theorem 3.1. *Let $1 < p < \infty$. If $f \in L^p(\mathbb{T})$, then $\|s_N(f) - f\|_{L^p(\mathbb{T})} \to 0$ as $N \to \infty$.*

The proof of this fact is quite involved when $p \neq 2$ and the $p = 2$ case is only easier due to the aforementioned Hilbert space structure, which we have not proven, so we omit the proof of this theorem altogether (see Katznelson for a proof) and instead investigate its consequences.

Recall that the L^p norms in a sense measure how large a function is. In particular, since $\mathbb{T}$ is compact and a finite measure space, the L^p norms on $\mathbb{T}$ are essentially measuring the vertical fluctuations of the function. The L^∞ norm measures the (almost everywhere) maximum of the function, while the L^1 norm measures the total mass of the function. For $1 < p < \infty$, the L^p norms measure a sort of weighted average of these quantities.

The fact that the theorem fails for exactly $p = 1$ and $p = \infty$ tells us that the outliers happen at the extreme cases. The Fourier partial sums may behave wildly enough that they keep on fluctuating in mass, and it is also possible that there are some spikes in the partial sums that never fully go away.

Remark 3.2. Beyond telling us in what ways the Fourier partial sums can behave, L^2 convergence is particularly useful (again due to its Hilbert space structure) since it allows us to swap integrals and sums for Fourier series, due to the continuity of the inner product.

3.2. Case $p = 1$.

In the case of L^1 we can give some more explicit bounds. We start off with the following theorem:

Theorem 3.3 (Uniform Boundedness Principle)**.** *Let X be a Banach space and Y be a normed vector space, and let $\mathcal{B}(X, Y)$ denote the space of bounded linear operators from X to Y equipped with the operator norm. Then given $\mathcal{F} \subseteq \mathcal{B}(X, Y)$, if $\sup_{T\in\mathcal{F}} \|Tx\|_Y < \infty$ for all $x \in X$, then $\sup_{T\in\mathcal{F}} \|T\|_{op} < \infty$.*

We give the following proof without the use of the Baire Category theorem due to Sokal.

Lemma 3.4. *Let $T \in \mathcal{B}(X, Y)$ as defined above. Then for any $x_0 \in X$ and $r > 0$ we have*

$$\sup_{x\in B(x_0,r)} \|Tx\|_Y \geq \|T\|_{op} r$$

Proof. For $\xi \in X$, we have by triangle inequality that

$$\max\{\|T(x_0 + \xi)\|_Y, \|T(x_0 - \xi)\|_Y\} \geq \frac{1}{2} (\|T(x_0 + \xi)\|_Y + \|T(x_0 - \xi)\|_Y) \geq \|T\xi\|_Y$$

Now we take a supremum over $\xi \in B(0, r)$ giving

$$\sup_{x\in B(x_0,r)} \|Tx\|_Y \geq \sup_{\xi\in B(0,r)} \|T\xi\|_Y = r \sup_{\xi/r\in B(0,1)} \|T(\xi/r)\|_Y = \|T\|_{op} r$$

□

Proof of Uniform Boundedness Principle. We prove by contrapositive. Suppose that $\sup_{T\in\mathcal{F}} \|T\|_{op} = \infty$ so we can choose $\{T_n\}_{n\in\mathbb{N}} \subseteq \mathcal{F}$ such that $\|T_n\|_{op} \geq 4^n$. Now let $x_0 = 0$, and use the above lemma to inductively choose $x_n \in X$ such that $\|x_n - x_{n-1}\|_X \leq 3^{-n}$ and $\|T_n x_n\|_Y \geq \frac{2}{3} 3^{-n} \|T_n\|_{op}$ (We applied the lemma to T_n and the point x_{n-1} with radius $\|x_n - x_{n-1}\|_X$.) $\{x_n\}$ is Cauchy, so it converges to some $x \in X$, since X is a Banach space, and we can see that $\|x - x_n\|_X \leq \frac{1}{2} 3^{-n}$. Thus

$$\|T_n x\|_Y \geq |\|T_n x_n\|_Y - \|T_n(x - x_n)\|_Y| \geq \frac{1}{6} 3^{-n} \|T_n\|_{op} \geq \frac{1}{6} (4/3)^n \to \infty$$

□

This lets us prove an equivalent condition for L^p convergence:

Lemma 3.5. *Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{T})$. Then $s_N(f) \to f$ in L^p norm iff there exists C_p independent of N such that*

$$\|s_N(f)\|_{L^p(\mathbb{T})} \leq C_p \|f\|_{L^p(\mathbb{T})}$$

Proof. ($\Longrightarrow$) We can check that $s_N : L^p(\mathbb{T}) \to L^p(\mathbb{T})$ is a bounded linear operator on f. Since $s_N(f) \to f$ in L^p, it follows $s_N(f)$ must be bounded in L^p norm, so $\sup_N \|s_N(f)\|_{L^p(\mathbb{T})} < \infty$. Then, by the uniform boundedness principle, we must have $\sup_N \|s_N\|_{op} < \infty$. Denote this quantity by C_p. We then have our desired inequality by definition of the operator norm.

($\Longleftarrow$) Now suppose the inequality is satisfied. Note that if g is a trigonometric polynomial then $s_N(g) = g$ for all sufficiently large N. Let $f \in L^p(\mathbb{T})$. By the density of trigonometric polynomials in $L^p(\mathbb{T})$, we can find a trigonometric polynomial g such that $\|f - g\|_{L^p(\mathbb{T})} \leq \varepsilon$. Then

$$\|s_N(f) - f\|_{L^p(\mathbb{T})} \leq \|s_N(f) - s_N(g)\|_{L^p(\mathbb{T})} + \|s_N(g) - g\|_{L^p(\mathbb{T})} + \|g - f\|_{L^p(\mathbb{T})} \leq (C_p + 1)\varepsilon$$

Thus $s_N(f) \to f$ in L^p norm. □

This tells us that the failure of L^p convergence when $p = 1$ or $p = \infty$ happens due to the lack of the boundedness of this partial sum operator. For the case of $p = 1$ we can approximate the growth of the operator norm of s_N.

Proposition 3.6. $\|s_N\|_{op} = \dfrac{4}{\pi^2} \log N + O(1)$

This means that in the worse case scenario, we would not be able find a finite constant C_p satisfying the above bound.

Definition 3.7 (Dirichlet kernel)**.** We define the Dirichlet kernel

$$D_N(t) := \sum_{n=-N}^{N} e^{int}$$

Note then that

$$\begin{aligned} s_N(f)(x) &= \sum_{n=-N}^{N} \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-int}\, dt \cdot e^{inx} \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(t) D_N(x - t)\, dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(x - t) D_N(t)\, dt \end{aligned}$$

This is called the convolution of f and D_N, denoted by $s_N(f) = f * D_N$.

We can also compute D_N as a geometric sum to get

$$D_N(t) = \frac{e^{-iNt} - e^{i(N+1)t}}{1 - e^{it}} = \frac{e^{-i(N+1/2)t} - e^{i(N+1/2)t}}{e^{-it/2} - e^{it/2}} = \frac{\sin((N + 1/2)t)}{\sin(t/2)}$$

Definition 3.8. We define the Lebesgue numbers

$$L_N := \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(t)|\, dt = \|D_N\|_{L^1(\mathbb{T})}$$

Note the last equality comes from the fact that the measure is translation invariant.

Proposition 3.9. $L_N = \dfrac{4}{\pi^2}\log N + O(1)$

Proof. We can approximate

$$\begin{aligned}
L_N &= \frac{1}{2\pi}\int_{-\pi}^{\pi}\left|\frac{\sin((N+1/2)t)}{\sin(t/2)}\right| dt \\
&= \frac{1}{\pi}\int_{0}^{\pi}\left|\frac{\sin((N+1/2)t)}{\sin(t/2)}\right| dt \\
&= \frac{1}{\pi}\int_{0}^{\pi}\left|\frac{\sin((N+1/2)t)}{t/2}\right| dt + O(1) \\
&= \frac{2}{\pi}\int_{0}^{(N+1/2)\pi}\left|\frac{\sin t}{t}\right| dt + O(1) \\
&= \frac{2}{\pi}\sum_{n=0}^{N-1}\int_{n\pi}^{(n+1)\pi}\left|\frac{\sin t}{t}\right| dt + \frac{2}{\pi}\int_{N\pi}^{(N+1/2)\pi}\left|\frac{\sin t}{t}\right| dt + O(1) \\
&= \frac{2}{\pi}\sum_{n=0}^{N-1}\int_{0}^{\pi}\frac{|\sin t|}{t+n\pi}\, dt + O(1) \\
&= \frac{2}{\pi}\int_{0}^{\pi}\sin t\sum_{n=0}^{N-1}\frac{1}{t+n\pi}\, dt + O(1) \\
&= \frac{2}{\pi^2}\int_{0}^{\pi}\sin t(\log N + O(1))\, dt + O(1) \\
&= \frac{4}{\pi^2}\log N + O(1)
\end{aligned}$$

From the second line to the third line, note that $|1/\sin(t/2)| - |1/(t/2)|$ is bounded on $(0,\pi]$, hence the $O(1)$ approximation. For the fifth line, the integral is again bounded. From the seventh line to the eighth line, we use the Euler approximation for harmonic sums. □

Finally, we conclude with a functional analysis fact.

Lemma 3.10. *Let $g \in L^1(\mathbb{T})$. The convolution operator $f \mapsto f * g$ is a continuous linear map with operator norm $\|g\|_{L^1(\mathbb{T})}$.*

From the definition of s_N and D_N this does not seem to be a farfetched conclusion. Unfortunately, the amount of definitions and setup required to prove this fact from just the content in 245A is too much to include here, so we take this fact for granted.

Proof of Proposition 3.6. We have $s_N(f) = f * D_N$, so $\|s_N\|_{op} = \|D_N\|_{L^1(\mathbb{T})} = L_N$. The approximation then follows. □

Remark 3.11. Here we only explored how the Fourier sums grow in norm, but it turns out they can behave very wildly pointwise. Kolmogorov showed in 1923 that there exists a L^1 function whose Fourier series diverged almost everywhere, and in 1926 improved that to divergence everywhere. This will come up again when we discuss the pointwise convergence of Fourier series at the end.

3.3. **Case $p = \infty$ and Gibb's Phenomenon.** We have actually already seen a counterexample to L^∞ convergence.

Proposition 3.12. *Let $f_n, f \in L^\infty(\mathbb{T})$ for $n \in \mathbb{N}$. If $f_n \to f$ in L^∞ norm then*

$$\lim_{n\to\infty}\int_{\mathbb{T}} f_n = \int_{\mathbb{T}} f$$

Proof. We have $|f_n - f| \leq \|f_n - f\|_{L^\infty(\mathbb{T})}$ almost everywhere, so

$$\left|\int_{\mathbb{T}} f_n - f\right| \leq \int_{\mathbb{T}} |f_n - f| \leq \int_{\mathbb{T}} \|f_n - f\|_{L^\infty(\mathbb{T})} = \|f_n - f\|_{L^\infty(\mathbb{T})}\mu(\mathbb{T}) \to 0$$

□

Note that L^∞ convergence is essentially the same as uniform convergence for finite measure spaces. Recall our example of the Fourier series x. The discontinuity causes the series to not converge uniformly, so when we tried to use term-by-term differentiation we got complete nonsense.

The lack of convergence manifests itself in a graphical way for functions with discontinuities, known as Gibb's phenomenon. It says that at points with jump discontinuities, the Fourier partial sums will essentially "overshoot," and these spikes are exactly what causes the failure of convergence in L^∞.

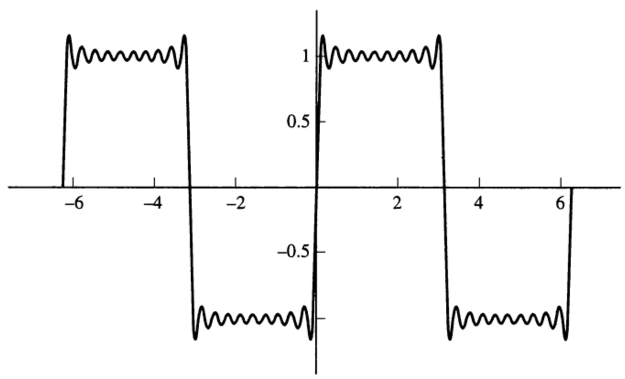

FIGURE 1. Gibb's Phenomenon for $f(x) = \text{sgn}(x)$. Taken from [3]

However, even if we don't get convergence L^∞, we might like to ask if we can say something about the growth rate of the Fourier partial sums like we did for L^1. In fact we have the following result:

Theorem 3.13. *Let $f \in C(\mathbb{T})$. Then*

$$\lim_{N\to\infty} \frac{\|s_N(f)\|_{L^\infty(\mathbb{T})}}{\log N} = 0$$

Proof. Note that because $\mathbb{T}$ is compact and $f \in C(\mathbb{T})$ we have $\|f\|_{L^\infty(\mathbb{T})} = \|f\|_\infty < \infty$ where $\|\cdot\|_\infty$ is the uniform norm, since the upper bound of the function is attained. Recall that we have

$$s_N(f)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)\frac{\sin((N+1/2)t)}{\sin(t/2)}\,dt$$

where we can shift the limits of integration by the invariance of measure. Then we have

$$\begin{aligned}|s_N(f)| &\leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x-t)|\left|\frac{\sin((N+1/2)t)}{\sin(t/2)}\right|\,dt \\ &\leq \frac{1}{2\pi}\|f\|_\infty \int_{-\pi}^{\pi}\left|\frac{\sin((N+1/2)t)}{\sin(t/2)}\right|\,dt\end{aligned}$$

Taking the L^∞ norm on both sides, and rewriting the uniform norm of f on the RHS, we get

$$\|s_N(f)\|_{L^\infty(\mathbb{T})} \leq \frac{1}{2\pi}\|f\|_{L^\infty(\mathbb{T})}\int_{-\pi}^{\pi}\left|\frac{\sin((N+1/2)t)}{\sin(t/2)}\right|\,dt$$

Now we can estimate the integral on the RHS the same way we did for L_N, except now we only need an upper bound. Note that if we choose $\alpha \in (0, 1/\pi]$, we have $\sin(t/2) \geq \alpha t$ on $[0, \pi]$. (You can check this by finding the minimum of $\sin(t/2)/t$.) Then, we have

$$\begin{aligned}
\int_{-\pi}^{\pi} \left| \frac{\sin((N+1/2)t)}{\sin(t/2)} \right| dt &\leq 2\int_0^{\pi} \left| \frac{\sin((N+1/2)t)}{\sin(t/2)} \right| dt \\
&\leq 2\int_0^{\pi} \frac{|\sin((N+1/2)t)|}{\alpha t}\, dt \\
&= \frac{2}{\alpha} \int_0^{(N+1/2)\pi} \frac{|\sin t|}{t}\, dt \\
&\leq \frac{2}{\alpha} \sum_{n=0}^{N} \int_{n\pi}^{(n+1)\pi} \frac{|\sin t|}{t}\, dt \\
&\leq \frac{2}{\alpha} \int_0^{\pi} \frac{\sin t}{t}\, dt + \sum_{n=1}^{N} \frac{2}{\alpha n \pi}
\end{aligned}$$

Since we can continuously extend to $\sin t/t$ to $[0, \pi]$, we have that the integral is finite. Now we use the fact that $\sum_1^N 1/n = \log N + O(1)$ to show that

$$\frac{\|s_N(f)\|_{L^\infty(\mathbb{T})}}{\log N} \leq \|f\|_{L^\infty(\mathbb{T})} \left(\frac{C}{\log N} + \frac{2}{\alpha\pi \log N} \sum_{n=1}^{N} \frac{1}{n} \right) \leq K\|f\|_{L^\infty(\mathbb{T})}$$

for some constant K. Now define $\Lambda_N(f) := s_N(f)/\log N$. Notice that Λ_N is a linear operator.

Since trigonometric polynomials are dense in the continuous functions equipped with the uniform norm (which is the same as the L^∞ norm here), for any $\varepsilon > 0$ we can find a trigonometric polynomial P such that $\|f - P\|_{L^\infty(\mathbb{T})} \leq \varepsilon$. Then

$$\begin{aligned}
\|\Lambda_N(f)\|_{L^\infty(\mathbb{T})} &\leq \|\Lambda_N(f-P)\|_{L^\infty(\mathbb{T})} + \|\Lambda_N(P)\|_{L^\infty(\mathbb{T})} \\
&\leq K\|f-P\|_{L^\infty(\mathbb{T})} + \|\Lambda_N(P)\|_{L^\infty(\mathbb{T})} \\
&\leq \varepsilon + \|\Lambda_N(P)\|_{L^\infty(\mathbb{T})}
\end{aligned}$$

But trigonometric polynomials are bounded on $\mathbb{T}$ (they are continuous), and for sufficiently large N we have $\Lambda_N(P) = s_N(P)/\log N = P/\log N$, so we must have $\|\Lambda_N(P)\|_{L^\infty(\mathbb{T})} \to 0$ as $N \to \infty$. Finally, this implies $\|\Lambda_N(f)\|_{L^\infty(\mathbb{T})} \to 0$ as $N \to \infty$ as desired. □

Remark 3.14. Note that continuity is important here so the function is bounded, but also because trigonometric polynomials are not dense in $L^\infty(\mathbb{T})$ since it is not separable.

Remark 3.15. Comparing the cases of L^1 and L^∞, we can only say s_N grows at most like $\log N$ in L^1 but we can say that s_N grows much slower than $\log N$ in L^∞.

4. Carleson-Hunt Theorem

Up until now, we have only discussed convergence in the L^p norms. This can tell us about how the partial Fourier sums oscillate, but there it does not tell us whether our Fourier series actually represents our function. One of the most important questions is whether we have pointwise convergence of Fourier series. There are various sufficient conditions that have been given since the development of Fourier series, but no one knew if this was true for a wide class of functions in general until Carleson and Hunt proved the following result in 1966 and 1967:

Theorem 4.1 (Carleson-Hunt)**.** *Let $1 < p \leq \infty$ and $f \in L^p(\mathbb{T})$. Then $s_N(f) \to f$ pointwise almost everywhere.*

This turns out to be much more difficult to prove than L^p convergence, but notably it fails for $p = 1$. This is again due to the fact that the Fourier partial sums of L^1 functions can diverge wildly, as was shown by Kolmogorov's counterexample.

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